

□ Python Must-Do Questions (for Interviews)

□ These are most common, pattern-based and cover basics □ OOP □ small utilities.

Basics & Syntax

1.
Write a Python program to swap two numbers without using a third variable.
2.
Check if a given number is prime.
3.
Find the factorial of a number using recursion.
4.
Reverse a string without using slicing (`[::-1]`).
5.
Count vowels and consonants in a string.
6.
Find the largest element in a list.
7.
Remove duplicates from a list.
8.
Merge two sorted lists into one sorted list.

Data Structures (Lists, Dicts, Sets, Tuples)

9.
Implement a stack using a list.
10.
Implement a queue using collections.deque.
11.
Count frequency of words in a string (use dict).
12.
Find the second largest element in a list.
13.
Flatten a nested list (example: [1,[2,[3]]] → [1,2,3]).
14.
Find common elements in two lists.
15.
Convert two lists into a dictionary (keys + values).

String/Regex

16.
Check if two strings are anagrams.
17.
Check if a string is palindrome.
18.
Find the first non-repeating character in a string.

19.
Extract email IDs from a string using regex.

OOP in Python

20.
Create a class BankAccount with deposit and withdraw methods.
21.
Implement inheritance with Animal → Dog.
22.
Demonstrate polymorphism using method overriding.
23.
Use @staticmethod and @classmethod with examples.

Pythonic / Advanced

24.
Use list comprehension to generate all even numbers from 1–100.
25.
Write a lambda function to sort a list of tuples by second element.
26.
Demonstrate *args and **kwargs usage.
27.
Implement exception handling for division by zero.

28.

Read/write from a file, and count the number of lines.

29.

Use map, filter, and reduce in examples.

30.

Demonstrate generator with yield keyword.

□ DSA Must-Do Questions (Python-friendly)

Arrays (Core must)

1.
Find the largest and smallest element in an array.
2.
Reverse an array in-place.
3.
Find the second largest element in an array.
4.
Move all zeros to the end of the array.
5.
Find the missing number in an array of 1..n.
6.
Find duplicates in an array.
7.
Find the intersection of two arrays.
8.
Sort an array of 0s, 1s, and 2s (Dutch National Flag).
9.
Maximum sum subarray (Kadane's Algorithm).
10.
Rotate an array by k steps.

Strings (Core must)

11.

Check if a string is palindrome.

12.

Reverse words in a string ("the sky is blue" → "blue is sky the").

13.

Check if two strings are anagrams.

14.

Find the first non-repeating character.

15.

Find the longest substring without repeating characters (sliding window).

16.

Implement substring search (naïve + mention KMP idea if you can).

17.

Count frequency of characters in a string.

Searching & Sorting

18.

Binary Search (iterative + recursive).

19.
Find first and last occurrence of an element in sorted array.
 20.
Search in rotated sorted array.
 21.
Merge two sorted arrays.
 22.
Implement quicksort OR mergesort (just know one).
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Hashing / Sets

23.
Two sum problem (check if any two numbers sum to target).
 24.
Subarray with given sum (use hashmap).
 25.
Longest consecutive sequence in an array.
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Two Pointers / Sliding Window

26.
Pair sum in sorted array (two pointer).

27.

Triplet sum in array.

28.

Maximum sum subarray of size k.

29.

Minimum window substring (harder, but good to know pattern).

Optional DP (only if time permits — interviewers like these sometimes)

30.

Longest Common Subsequence (LCS).

31.

Longest Palindromic Subsequence.

32.

Edit Distance.

□ If time is tight, skip DP. Just know how to explain recurrence + $dp[i][j]$ meaning so you don't blank.

□ Core CS Must-Revise Qs

DBMS (high priority)

1.
Explain Primary Key, Foreign Key, Candidate Key, Composite Key with examples.
2.
Difference between DELETE, TRUNCATE, DROP.
3.
Types of JOINS (INNER, LEFT, RIGHT, FULL).
4.
What is Normalization? Explain 1NF, 2NF, 3NF with example.
5.
What is a Transaction? Explain ACID properties.
6.
What is Indexing? Advantages and disadvantages.
7.
Difference between SQL vs NoSQL databases.
8.
Write an SQL query:
 - Find 2nd highest salary from Employee table.

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Find all employees who work in 'HR' department.

Operating System (high priority)

9.

Difference between Process vs Thread.

10.

What is Scheduling? Explain FCFS, SJF, Round Robin, Priority.

11.

What is Deadlock? Give 4 conditions + prevention.

12.

What is Paging vs Segmentation?

13.

What is Virtual Memory?

14.

Difference between Multithreading vs Multiprocessing.

15.

What is a Context Switch?

Computer Networks (medium priority)

16.
Difference between OSI model vs TCP/IP model.

17.
Explain TCP vs UDP with real-life examples.

18.
What is the difference between HTTP vs HTTPS?

19.
What is DNS? How does it work?

20.
What is an IP Address? Difference between IPv4 and IPv6.

21.
What is Subnetting (basic idea only)?

OOPS (if company asks Python/Java basics)

22.
Explain OOP Principles: Encapsulation, Abstraction, Inheritance, Polymorphism.

23.
What is the difference between Class vs Object?

24.
What is Method Overloading vs Method Overriding?

25.
What is the difference between Abstract class vs Interface?
26.
Example of constructor in Python/Java.
27.
Difference between static vs instance methods.
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Extra / CS Fundamentals (if asked)

28.
What is Compiler vs Interpreter?
29.
What is Time Complexity vs Space Complexity? (Big-O basics)
30.
What is Hashing? Where is it used?
31.
Difference between Stack vs Queue.
32.
Difference between Linked List vs Array.